

JPE Replication Report

PUBLISHED

October 1, 2026

1 Paper Identification

ID	20251367
Author	Rivera
Title	Market Response to Racial Uprisings
Iteration	2

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. Your package ran for 7 days on our system. Given you specified a 32GB RAM requirement, we had a 64GB RAM machine. Please add a warning to the readme. You should try to measure actual RAM consumption. See [here](#) for an example for how to achieve that.
2. While most of the exhibits were identical, we have 3 figures where this is not the case. Can you check those?
3. Tables A.10 and A.11 are hard coded. Where do those numbers come from? If you created them with code, this would be easy to document.

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☒ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☒ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☒ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☒ License for Code (Optional, but recommended)
- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.

- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
CRSP Daily/Monthly Security Files	no	yes	yes	yes	yes	yes
WRDS Beta Suite (CAPM, FF4)	no	yes	yes	yes	yes	yes
Compustat Fundamentals (Annual, Quarterly)	no	yes	yes	yes	yes	yes
Fama-French Monthly Factors	no	yes	yes	yes	yes	yes
I/B/E/S Analyst Consensus Recommendations	no	yes	yes	yes	yes	yes
Bloomberg Government/Police Client Data	no	yes	-	yes	yes	yes
Police1 Vendor Rosters (PD Atlas)	no	yes	-	yes	yes	yes
EDGAR 10-K Term Frequencies	yes	no	-	yes	yes	yes
Producer Price Index (PPI)	yes	no	yes	yes	yes	yes
Private Security Firm Classifications	no	yes	-	yes	yes	yes
ISC Exhibitor Lists	no	yes	-	yes	yes	yes
UCR Crime Statistics	yes	no	yes	yes	yes	yes

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
LEOKA (Officers Killed & Assaulted)	yes	no	yes	yes	yes	yes
LEMAS (Law Enforcement Management)	yes	no	yes	yes	yes	yes
Mapping Police Violence	yes	no	yes	yes	yes	yes
GDELT Protest Media Coverage	yes	no	yes	yes	yes	yes
NOAA Weather Data	yes	no	yes	yes	yes	yes
Masten & Poirier (2020) Replication Files	no	yes	-	yes	yes	yes
Product/Service Categorization	no	yes	-	yes	yes	yes
Twitter/X Daily Engagement Data	no	no	-	yes	yes	yes
ISS CEO Demographics	no	yes	yes	yes	yes	yes
U.S. Census Bureau (ACS 5-Year, 2009–2013)	yes	no	yes	yes	yes	yes
U.S. Census Bureau (TIGER/Line)	yes	no	yes	yes	yes	yes

5.1.1 CRSP Daily/Monthly Security Files

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided or obtained through restricted access
- ☒ DOI or URL is provided, and works: <https://wrds-www.wharton.upenn.edu>
- ☒ Access conditions are described:
 - A WRDS subscription is required.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

CRSP Daily/Monthly Security Files. Source: WRDS. Stored in [raw_data/wrds/](#). Access requires a WRDS subscription: <https://wrds-www.wharton.upenn.edu>.

5.1.2 WRDS Beta Suite (CAPM, FF4)

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided or obtained through restricted access
- ☒ DOI or URL is provided, and works: <https://wrds-www.wharton.upenn.edu>
- ☒ Access conditions are described:
 - The same WRDS subscription requirement applies.

- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

WRDS Beta Suite, including CAPM and Fama-French four-factor estimates. Source: WRDS. Stored in [raw_data/wrds/](#). Access requires a WRDS subscription.

5.1.3 Compustat Fundamentals (Annual, Quarterly)

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided or obtained through restricted access
- ☒ DOI or URL is provided, and works: <https://wrds-www.wharton.upenn.edu>
- ☒ Access conditions are described:
 - The same WRDS subscription requirement applies.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

Compustat Fundamentals, annual and quarterly files. Source: WRDS. Stored in [raw_data/wrds/](#). Access requires a WRDS subscription.

5.1.4 Fama-French Monthly Factors

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided or obtained through restricted access
- ☒ DOI or URL is provided, and works: <https://wrds-www.wharton.upenn.edu>
- ☒ Access conditions are described:
 - The README identifies the same WRDS subscription requirement.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

Fama-French Monthly Factors. Sources: WRDS and Ken French. Stored in [raw_data/wrds/](#). The access entry refers to the WRDS subscription requirement.

5.1.5 I/B/E/S Analyst Consensus Recommendations

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided or obtained through restricted access
- ☒ DOI or URL is provided, and works: <https://wrds-www.wharton.upenn.edu>
- ☒ Access conditions are described:
 - The same WRDS subscription requirement applies.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

I/B/E/S Analyst Consensus Recommendations. Source: WRDS. Stored in [raw_data/ibes/](#). Access requires a WRDS subscription.

5.1.6 Bloomberg Government/Police Client Data

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided or obtained through restricted access
- ☐ DOI or URL is provided
 - No dataset-specific DOI or URL is provided.
- ☒ Access conditions are described:

- Bloomberg Terminal access is required to reproduce the original extraction.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

Bloomberg Government/Police Client Data. Source: Bloomberg Terminal. Stored in [raw_data/bloomberg/](#). Terminal access is required for the original extraction.

5.1.7 Police1 Vendor Rosters (PD Atlas)

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided or collected
- ☐ DOI or URL is provided
 - No DOI or URL is provided.
- ☒ Access conditions are described:
 - The records were collected by the authors from Police1.com conference directories.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

Police1 Vendor Rosters, also described as PD Atlas. Source: Police1.com conference directories. Stored in [raw_data/rosters/](#). Collected by the authors.

5.1.8 EDGAR 10-K Term Frequencies

- ☒ Dataset is publicly accessible
- ☐ Dataset is PRIVATELY provided
- ☐ DOI or URL is provided
 - No specific URL is listed in the dataset table.
- ☒ Access conditions are described:
 - The underlying SEC EDGAR filings are public; term extraction was performed by the authors.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

EDGAR 10-K Term Frequencies. Sources: SEC EDGAR filings and the authors' text analysis. Stored in [raw_data/edgar/](#). The filings are public, while the term-frequency extraction was conducted by the authors.

5.1.9 Producer Price Index (PPI)

- ☒ Dataset is publicly accessible
- ☐ Dataset is PRIVATELY provided
- ☒ DOI or URL is provided, and works: <https://fred.stlouisfed.org/series/WPSFD41312>
- ☒ Access conditions are described:
 - The data are publicly available through FRED.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

Producer Price Index. Source: FRED. Stored in [raw_data/fred/](#). Publicly available at <https://fred.stlouisfed.org/series/WPSFD41312>.

5.1.10 Private Security Firm Classifications

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided or constructed

- ☐ DOI or URL is provided
 - No DOI or URL is provided.
- ☒ Access conditions are described:
 - The classifications were hand-coded by research assistants.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

Private Security Firm Classifications. Source: author hand-coding. Stored in [raw_data/private_security/](#). The classifications were hand-coded by research assistants.

5.1.11 ISC Exhibitor Lists

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided or collected
- ☐ DOI or URL is provided
 - No DOI or URL is provided.
- ☒ Access conditions are described:
 - The lists were collected by the authors from ISC West and ISC East conferences.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

ISC Exhibitor Lists. Source: ISC West and ISC East conferences. Stored in [raw_data/private_security/](#). Collected by the authors.

5.1.12 UCR Crime Statistics

- ☒ Dataset is publicly accessible
- ☐ Dataset is PRIVATELY provided
- ☒ DOI or URL is provided, and works: <https://www.fbi.gov/how-we-can-help-you/more-fbi-services-and-information/ucr>
- ☒ Access conditions are described:
 - The data are publicly available through the FBI Uniform Crime Reporting Program.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

UCR Crime Statistics. Source: FBI Uniform Crime Reporting Program. Stored in [raw_data/protests/](#). Publicly available through the FBI UCR website.

5.1.13 LEOKA (Officers Killed & Assaulted)

- ☒ Dataset is publicly accessible
- ☐ Dataset is PRIVATELY provided
- ☒ DOI or URL is provided, and works: <https://www.fbi.gov/how-we-can-help-you/more-fbi-services-and-information/ucr>
- ☒ Access conditions are described:
 - The access entry states "Public: same," referring to the FBI UCR link above.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

LEOKA, Law Enforcement Officers Killed and Assaulted. Source: FBI. Stored in [raw_data/protests/](#). Publicly available through the FBI UCR website.

5.1.14 LEMAS (Law Enforcement Management)

- ☒ Dataset is publicly accessible
- ☐ Dataset is PRIVATELY provided
- ☒ DOI or URL is provided, and works: <https://bjs.ojp.gov/data-collection/law-enforcement-management-and-administrative-statistics-lemas>
- ☒ Access conditions are described:
 - The data are publicly available through the Bureau of Justice Statistics.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

Law Enforcement Management and Administrative Statistics, LEMAS. Source: Bureau of Justice Statistics. Stored in [raw_data/protests/](#). Publicly available from the BJS LEMAS page.

5.1.15 Mapping Police Violence

- ☒ Dataset is publicly accessible
- ☐ Dataset is PRIVATELY provided
- ☒ DOI or URL is provided, and works: <https://mappingpoliceviolence.us>
- ☒ Access conditions are described:
 - The data are described as publicly available.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

Mapping Police Violence. Source: MPV. Stored in [raw_data/protests/](#). Publicly available at <https://mappingpoliceviolence.us>.

5.1.16 GDELT Protest Media Coverage

- ☒ Dataset is publicly accessible
- ☐ Dataset is PRIVATELY provided
- ☒ DOI or URL is provided, and works: <https://www.gdeltproject.org>
- ☒ Access conditions are described:
 - The data are described as publicly available through the GDELT Project.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

GDELT Protest Media Coverage. Source: GDELT Project. Stored in [raw_data/protests/](#). Publicly available at <https://www.gdeltproject.org>.

5.1.17 NOAA Weather Data

- ☒ Dataset is publicly accessible
- ☐ Dataset is PRIVATELY provided
- ☒ DOI or URL is provided, and works: <https://www.nci.noaa.gov/products/land-based-station/global-historical-climatology-network-daily>
- ☒ Access conditions are described:
 - The data are publicly available through NOAA's Global Historical Climatology Network Daily product.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

NOAA Weather Data. Source: NOAA GHCN-Daily. Stored in [raw_data/protests/get_weather/](#). Publicly available through NOAA's Global Historical Climatology Network Daily page.

5.1.18 Masten & Poirier (2020) Replication Files

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided or available upon request
- ☐ DOI or URL is provided
 - No DOI or URL is provided.
- ☒ Access conditions are described:
 - The replication files are available from the original authors.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

Masten and Poirier (2020) Replication Files. Source: authors of the cited paper. Stored in [raw_data/protests/MastenPoirier_2020_FAS_replication_files/](#). Available from the original authors.

5.1.19 Product/Service Categorization

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided or collected
- ☐ DOI or URL is provided
 - No DOI or URL is provided.
- ☒ Access conditions are described:
 - The categorization is based on author-collected survey responses gathered in May 2022.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

Product/Service Categorization. Source: author-collected survey responses. Stored in [raw_data/categorization/](#). Collected by the authors in May 2022.

5.1.20 Twitter/X Daily Engagement Data

- ☐ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided
- ☐ DOI or URL is provided
 - The table says "See citation below," but no URL is visible in the supplied screenshot.
- ☒ Access conditions are described:
 - Access information is deferred to the associated citation.
- ☒ Data are not directly cited here, but a working paper, article, or other document is cited
 - The source is identified as Dunivin et al. (2022).
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

Twitter/X Daily Engagement Data. Source: Dunivin et al. (2022). Stored in [raw_data/twitter/](#). The README directs readers to the corresponding citation for access information.

5.1.21 ISS CEO Demographics

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided or obtained through restricted access
- ☒ DOI or URL is provided, and works: <https://www.issgovernance.com>
- ☒ Access conditions are described:
 - A subscription to Institutional Shareholder Services is required.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

ISS CEO Demographics. Source: Institutional Shareholder Services. Stored as `raw_data/rosters/ISS_table_ceo.csv.gz`. Access requires an ISS subscription: <https://www.issgovernance.com>.

5.1.22 U.S. Census Bureau (ACS 5-Year, 2009–2013)

- ☒ Dataset is publicly accessible
- ☐ Dataset is PRIVATELY provided
- ☒ DOI or URL is provided, and works: <https://data.census.gov>
- ☒ Access conditions are described:
 - The American Community Survey data are publicly available through the U.S. Census Bureau.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

U.S. Census Bureau ACS five-year data, 2009–2013. Source: American Community Survey. Stored as `raw_data/protests/county_covariates2020.csv.gz` and `raw_data/protests/census_state.csv.gz`. Publicly available at <https://data.census.gov>.

5.1.23 U.S. Census Bureau (TIGER/Line)

- ☒ Dataset is publicly accessible
- ☐ Dataset is PRIVATELY provided
- ☒ DOI or URL is provided, and works: <https://www2.census.gov/geo/tiger/>
- ☒ Access conditions are described:
 - The geographic files are publicly available from the U.S. Census Bureau.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

U.S. Census Bureau TIGER/Line geographic files. Source: Census Bureau geography. Stored in `raw_data/protests/get_weather/tl_2020_us_county/`. Publicly available at <https://www2.census.gov/geo/tiger/>.

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☒ Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
 - Stata 16 or higher (StataSE or StataNow recommended)
 - R 4.0 or higher
 - Python 3.8 or higher, reachable from Stata
- ☒ Computational Requirements specified as follows:
 - **RAM:** 32 GB minimum (some intermediate daily files exceed 4 GB)
 - **Disk:** ~50 GB for full package including intermediate data
 - **OS:** Tested on macOS. Should work on Linux/Windows with appropriate Stata/R installations.
 - **Machine used for replication:** Apple M3 Ultra, 256 GB RAM, macOS 26.3, Stata 19 (StataNow), R 4.5.2. Total runtime: approximately 90 hours.
- ☒ Time Requirements specified as follows:
 - 90 hours.
- ☒ Requirements are complete.

7 Analyzing Data and Code Files

{REPLICATOR} Here I provide you with a list of what I identified as *data files* based on file endings. Please check the list and amend if there are obvious mistakes. Watch out for Personally Identifiable Information (PII)!

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary:

- Data files with PII indicators: 131
- Variables flagged in data: 2713
- Code files with PII references: 98
- PII references in code: 923

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	18_APR_2022_PN_CatTermFreq.csv.gz	2	name, loc, location
Data	20_APR_2022_PN_CatTermFreq_Crime.csv.gz	6	name, loc, location, son, lon, sex
Data	21_SEP_2024_AW_CatTermFreq_v3.csv.gz	2	name, loc, location
Data	38651-0001-Data.csv.gz	14	name, city, zip, sex, child, lat, loc, house, community, second
Data	38651-0001-Data.dta	119	name, city, zip, son, lat, sex, address, child, community, school, loc, social, location, house, network, census, second
Data	ISS_table_ceo.csv.gz	2	son, country
Data	MPVDatasetDownload_feb2024.csv.gz	27	name, gender, url, address, street, city, zip, lat, son, census, house, lon, district
Data	MPVDatasetDownload_feb2024.xlsx	27	name, gender, url, address, street, city, zip, lat, son, census, house, lon, district
Data	MPV_GDELT_ORI_events_strict.csv.gz	18	country, lname, name, lat, lon, url
Data	PD_atlas_rosters.csv.gz	1	name
Data	PD_atlas_rosters.xlsx	1	name
Data	PD_atlas_rosters_Master_IACP.csv.gz	7	name, address, city, zip, phone, email
Data	black_mpv_xy_summer2020.csv.gz	98	lat, lon

File Type	File	Variables/References	PII Categories
Data	black_mpv_xy_summer2020.dta	98	lat, lon
Data	category_res_0.csv.gz	1	name
Data	category_res_0.dta	1	name
Data	category_res_1.csv.gz	1	name
Data	category_res_1.dta	1	name
Data	category_res_2.csv.gz	1	name
Data	category_res_2.dta	1	name

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-09-14T17:20:13.346

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any windows filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that STATA allows / to be a filepath separator on a windows platform, hence this is a requirement for all STATA applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (C:\ etc)
120	11	35	1	0

7.1.3 Duplicate Files Report

We found the following duplicate files:

Filename	Size (MB)	Checksum (MD5)
cleaned_data/SDID/SDIDstrong_dfpooled_placebo_virtra_PsumAR.csv.gz	0.0	5d839debc0ac9b6cbb19aaa05306e379b6cc5f86
cleaned_data/SDID/SDIDstrong_dfpooled_placebo_smithwesson_PsumAR.csv.gz	0.0	35c6c940e896c78c6cea63152de8636ff6687bab
cleaned_data/SDID/SDIDstrong_dfpooled_placebo_vista_PsumAR.csv.gz	0.0	e9adb31ff237328d919722dd5c6f145fb9675b56
cleaned_data/SDID/SDIDstrong_dfpooled_placebo_PsumAR.csv.gz	0.0	bc33a482d987ab7c587325ab82ccbe34376d2a92
cleaned_data/SDID/SDIDstrong_dfpooled_ADL_placebo_APsumAR.csv.gz	0.0	421c43b097c6b9628b38be22ebfec4405190385a
cleaned_data/SDID/SDIDweak_dfpooled_ADL_placebo_APsumAR.csv.gz	0.0	5599c0d6e6ec751f81a09b54e8e4bd730ebdeb9c
cleaned_data/SDID/SDIDweak_dfpooled_placebo_PsumAR.csv.gz	0.0	f31dd6f04925fa5ca60f747593331e59996e9dec
cleaned_data/SDID/SDIDstrong_dfpooled_ADL_placebo_APsumAR.dta	0.03	5ff72d789982e94f29c416d5fbacae82ac43ded7
cleaned_data/SDID/SDIDstrong_dfpooled_placebo_PsumAR.dta	0.03	80bad477311ec7e48bf860a636ea4373aa5eaac5
cleaned_data/SDID/SDIDstrong_dfpooled_placebo_smithwesson_PsumAR.dta	0.03	15043e5509f256f3e220d50638cf6d372f485ebc
cleaned_data/SDID/SDIDstrong_dfpooled_placebo_virtra_PsumAR.dta	0.03	1affa18f589600e119426b3af0a9ca6480f933df
cleaned_data/SDID/SDIDstrong_dfpooled_placebo_vista_PsumAR.dta	0.03	908e0792016e5bca29e964d30d0f7cbc0f5ce307

Filename	Size (MB)	Checksum (MD5)
cleaned_data/SDID/SDIDweak_dfpooled_ADL_placebo_APsumAR.dta	0.03	9ba6ff887306abb271725142a86a3a58a1ea177b
cleaned_data/SDID/SDIDweak_dfpooled_placebo_PsumAR.dta	0.03	7aae792b4ed5777ae85ddfdbd6aa744f3c8464aad

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
cleaned_data/returns_moredays_AsianCEO.csv.gz	111.42
raw_data/wrds/securitymonthly.csv.gz	117.32
cleaned_data/intermediate_daily.csv.gz	127.85
raw_data/protests/get_weather/tl_2020_us_county/tl_2020_us_county.shp	132.67
raw_data/wrds/fundamentals_yr2023.csv.gz	153.8
cleaned_data/returns_moredays_privatesecurity_isc.dta	154.18
cleaned_data/returns_moredays_privatesecurity.csv.gz	161.31
cleaned_data/returns_moredays.dta	198.82
cleaned_data/returns_moredays_capm.csv	255.54
raw_data/protests/get_weather/output/county_weather_stations.csv	262.92
cleaned_data/returns_moredays_placebo_date.csv	271.86
cleaned_data/returns_moredays.csv	271.91
raw_data/wrds/fundamentalsquarterly.csv.gz	274.26
cleaned_data/returns_moredays_BlackCEO.dta	277.35
cleaned_data/returns_moredays_ADL_placebos.csv	294.12
cleaned_data/returns_daily_category_fullV4.csv	303.66
raw_data/protests/offenses_known_monthly_2019.dta	317.25
cleaned_data/intermediate_daily_capm.csv.gz	326.74
cleaned_data/returns_moredays_HispCEO.dta	394.65
cleaned_data/returns_moredays_HispanicCEO.dta	394.65
cleaned_data/returns_moredays_placebos.csv	400.5
cleaned_data/returns_moredays_BlackCEO.csv	439.33
cleaned_data/returns_moredays_AsianCEO.dta	466.11
cleaned_data/returns_moredays_ADL_placebos_BlackCEO.csv	467.09
cleaned_data/returns_moredays_HispCEO.csv	626.97

Filename	Size (MB)
cleaned_data/returns_moredays_HispanicCEO.csv	626.97
cleaned_data/returns_moredays_placebos_BlackCEO.csv	641.99
cleaned_data/returns_moredays_ADL_placebos_HispanicCEO.csv	647.16
raw_data/protests/weather.dta	665.34
cleaned_data/returns_moredays_privatesecurity.dta	719.18
cleaned_data/returns_moredays_AsianCEO.csv	740.29
cleaned_data/returns_moredays_ADL_placebos_AsianCEO.csv	768.43
cleaned_data/intermediate_daily_HispCEO.csv.gz	804.31
raw_data/wrds/securitydaily.csv.gz	826.77
cleaned_data/intermediate_daily_BlackCEO.csv.gz	844.61
cleaned_data/returns_moredays_placebos_HispanicCEO.csv	930.67
raw_data/wrds/betasuitedaily_capm.csv.gz	976.88
cleaned_data/returns_moredays_placebos_AsianCEO.csv	1058.53
cleaned_data/returns_moredays_privatesecurity.csv	1095.59
cleaned_data/intermediate_daily_HispanicCEO.csv.gz	1115.64
cleaned_data/returns_moredays_ADL_placebos_privatesecurity.csv	1190.15
cleaned_data/intermediate_daily_AsianCEO.csv.gz	1200.0
raw_data/protests/leoka_yearly_1960_2022.dta	1227.09
cleaned_data/intermediate_daily.dta	1341.89
raw_data/wrds/betasuitedaily_4factors.csv.gz	1347.29
cleaned_data/returns_moredays_placebos_privatesecurity.csv	1638.49
cleaned_data/intermediate_daily_capm.dta	1716.58
cleaned_data/intermediate_daily_HispCEO.dta	3435.36
cleaned_data/intermediate_daily_BlackCEO.dta	3581.14
cleaned_data/intermediate_daily_HispanicCEO.dta	4723.37
cleaned_data/intermediate_daily_AsianCEO.dta	5085.42

7.2 Code description

cloc		github.com/AIDanial/cloc v 2.06 T=1.22 s (122.7 files/s, 690319.0 lines/s)		
Language	files	blank	comment	code
CSV	21	0	0	688378
Text	5	19	0	126038
Markdown	2	627	0	11557

Language	files	blank	comment	code
Stata	65	1904	177	9502
R	39	689	732	2475
XML	2	3	0	1371
TeX	15	10	6	421
Python	1	23	24	124
---	---	---	---	---
SUM:	150	3275	939	839866

- ☒ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

7.3 Computing Environment of the Replicator

- Nuvolos: 64 GB, 16 CU.
- Stata/MP 19.5
- R 4.6.0
- Python 3.12.12

8 Replication

- Downloaded code and data from dropbox to the Nuvolos instance.
- Ran code as per README, but I hit some errors, see below.
- I had to give R a writable user library (this is due to the sandbox). Had to run: `bash mkdir -p ~/R/library export R_LIBS_USER=~R/library R -e '.libPaths()' R_LIBS_USER=~R/library Rscript scripts/00_setup.R`
- Due to sandbox restrictions I had to run: `bash curl -Ls https://micro.mamba.pm/api/micromamba/linux-64/latest | tar -xvj bin/micromamba ./bin/micromamba create -p ~/r-geo -c conda-forge r-base=4.3 r-sf r-tidyverse r-data.table r-dplyr r-tidyr r-lubridate r-readxl r-haven r-xtable r-remotes r-mvtnorm r-iridis r-ggextra r-janitor r-fedmatch r-stringdist -y`
- And then: `bash ~/r-geo/bin/Rscript scripts/00_setup.R export PATH="$HOME/r-geo/bin:$PATH" export R_LIBS_USER=""`
- I had to install `R.utils`:
`bash R_LIBS_USER="" ~/r-geo/bin/Rscript -e 'install.packages("R.utils", repos="https://cloud.r-project.org")'`
- Ran until completion from main script.
- All outputs were reproduced, some with discrepancies.

9 Findings

9.1 Data Preparation Code

- Program `scripts/0a_clean_roster.do` ran without error.
- Program `scripts/0b_clean_classification.do` ran without error.

9.2 Tables and Figures

- Figure 9A: Paper version looks different from the one generated by programs:

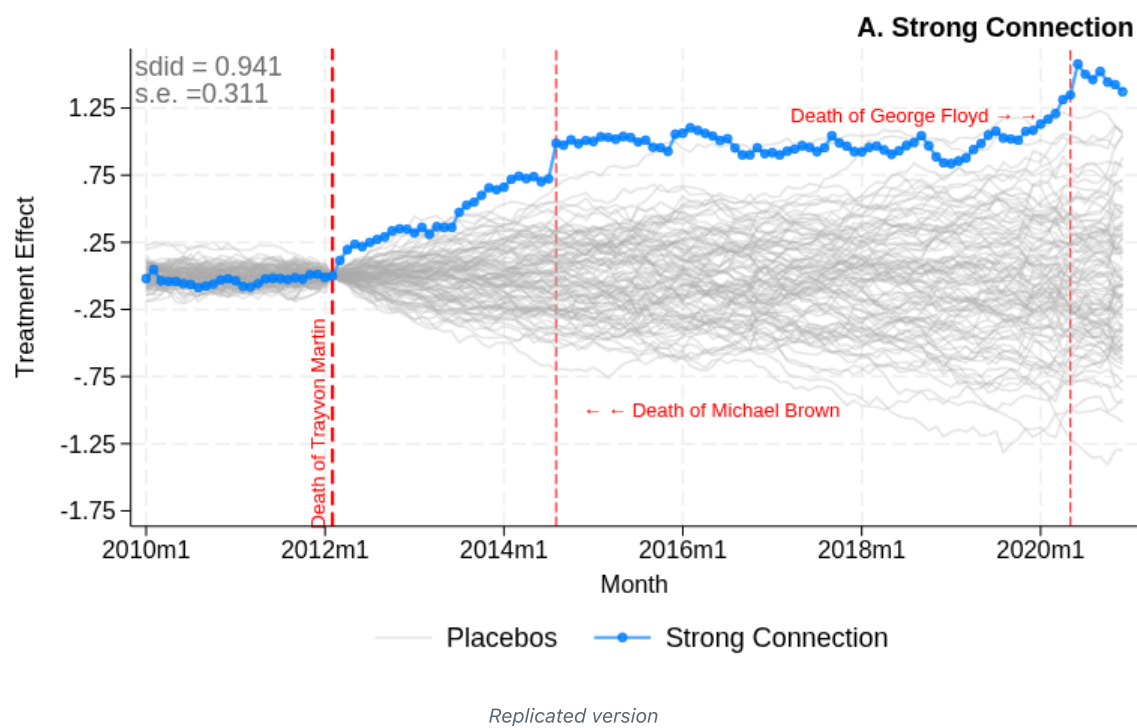
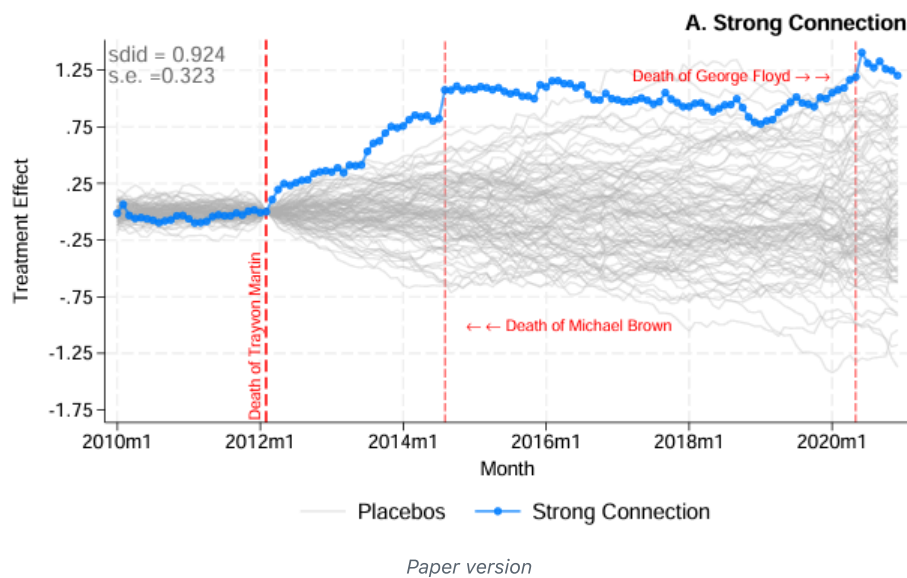


Figure 1: Figure 9A

- Figure 9B: Paper version looks different from the one generated by programs:

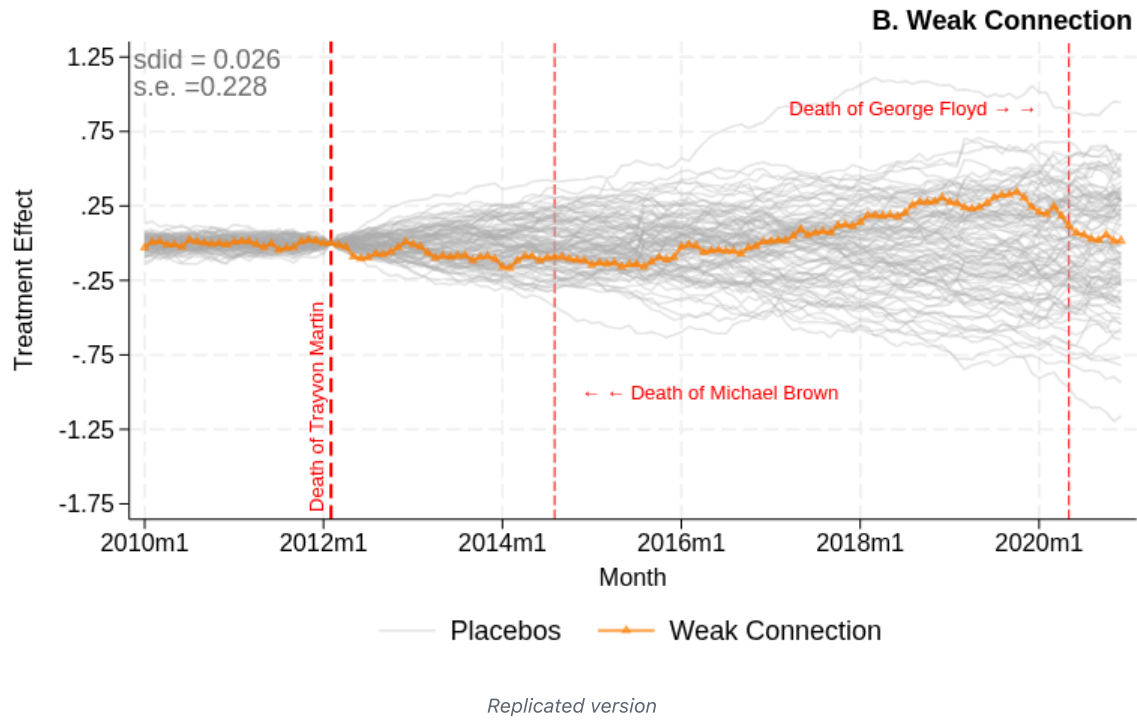
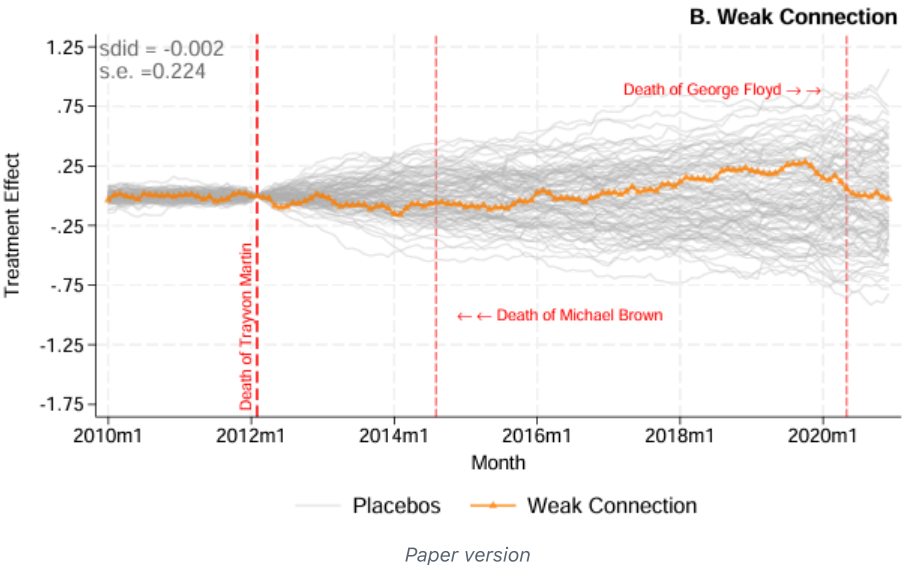


Figure 2: Figure 9B

- Figure A.12A: Paper version looks different from the one generated by programs:

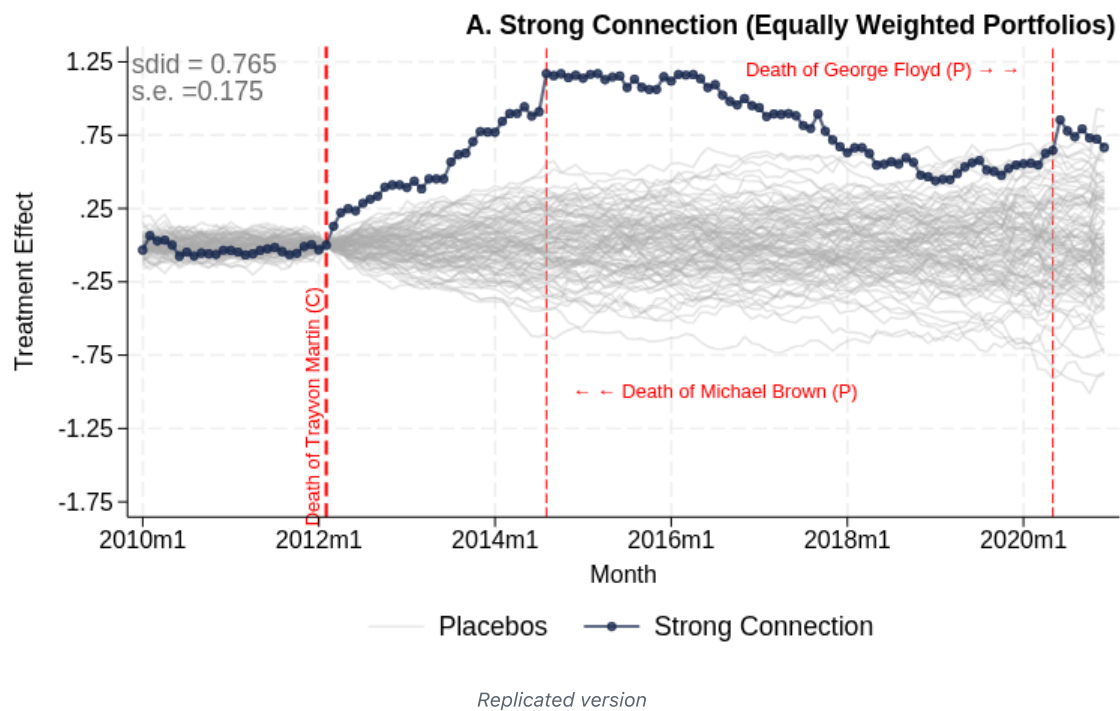
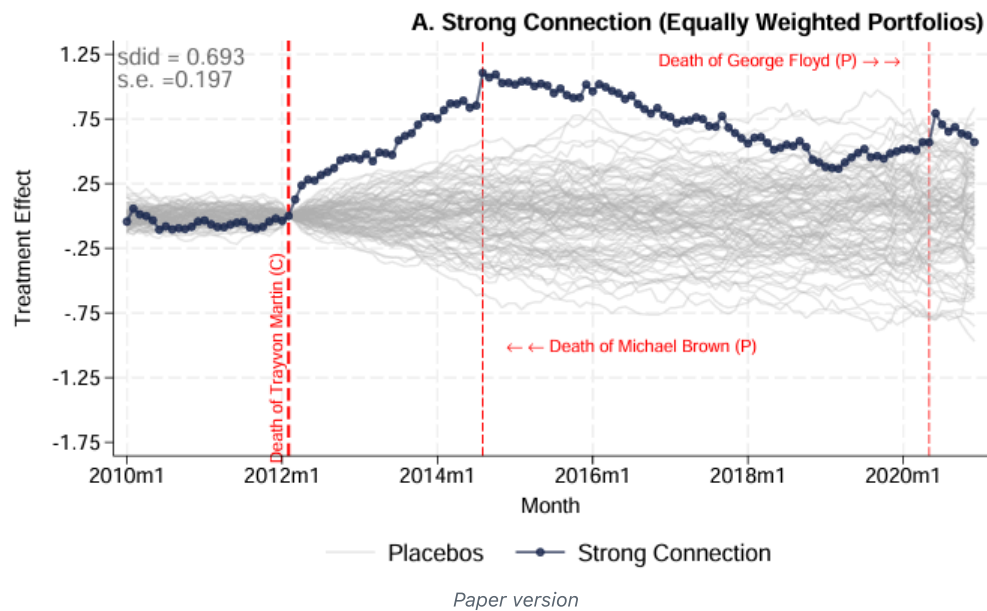


Figure 3: Figure A.12A

- Figure A.12B: Paper version looks different from the one generated by programs:

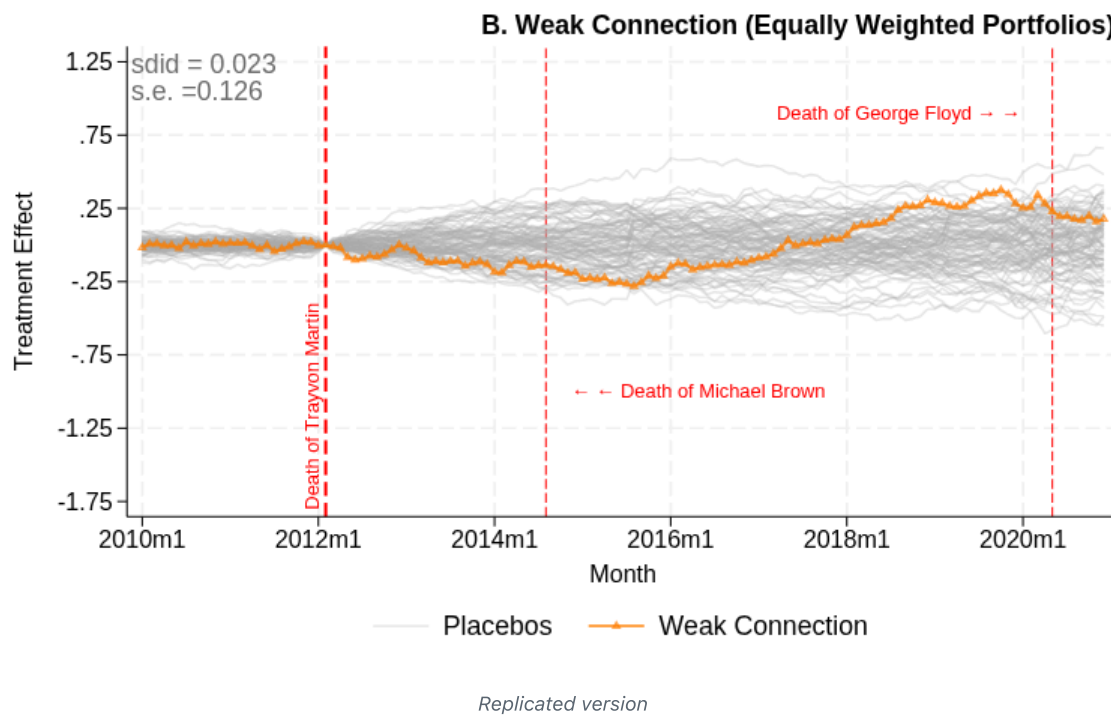
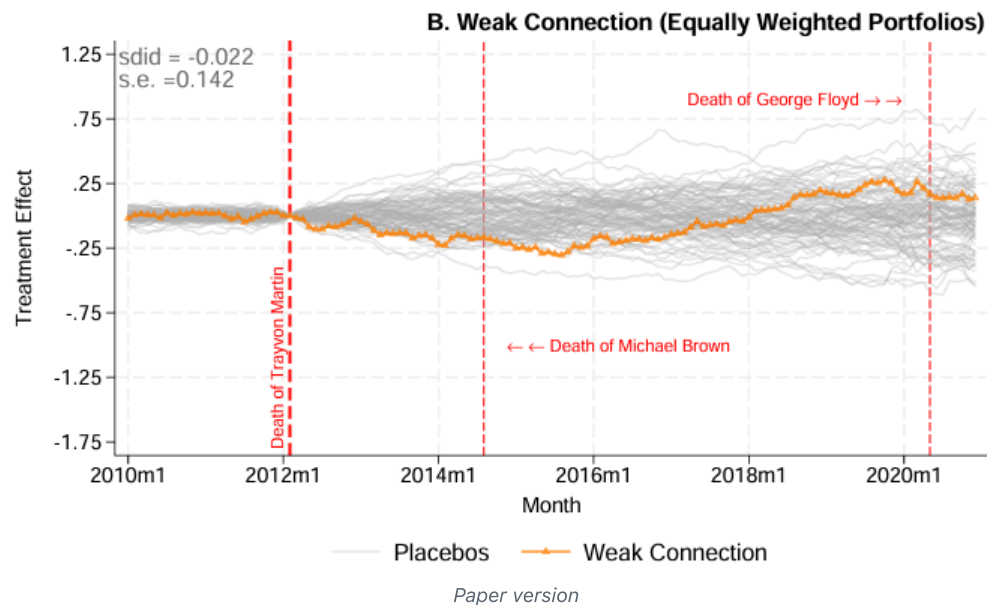


Figure 4: Figure A.12B

- Figure A.13: Paper version looks different from the one generated by programs:

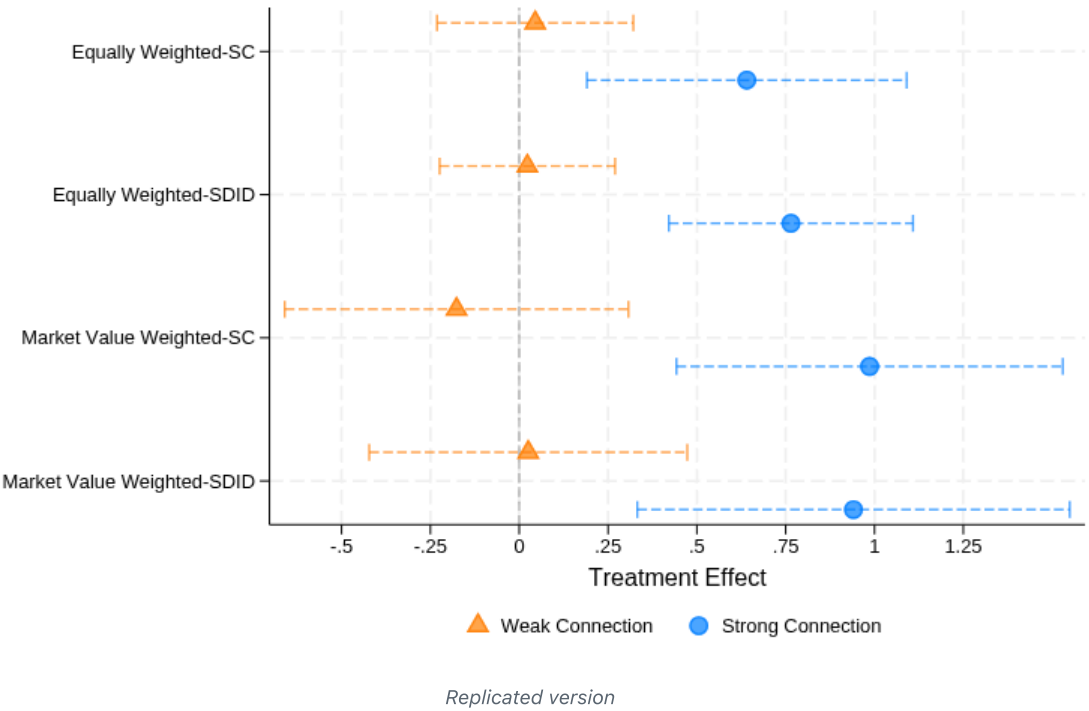
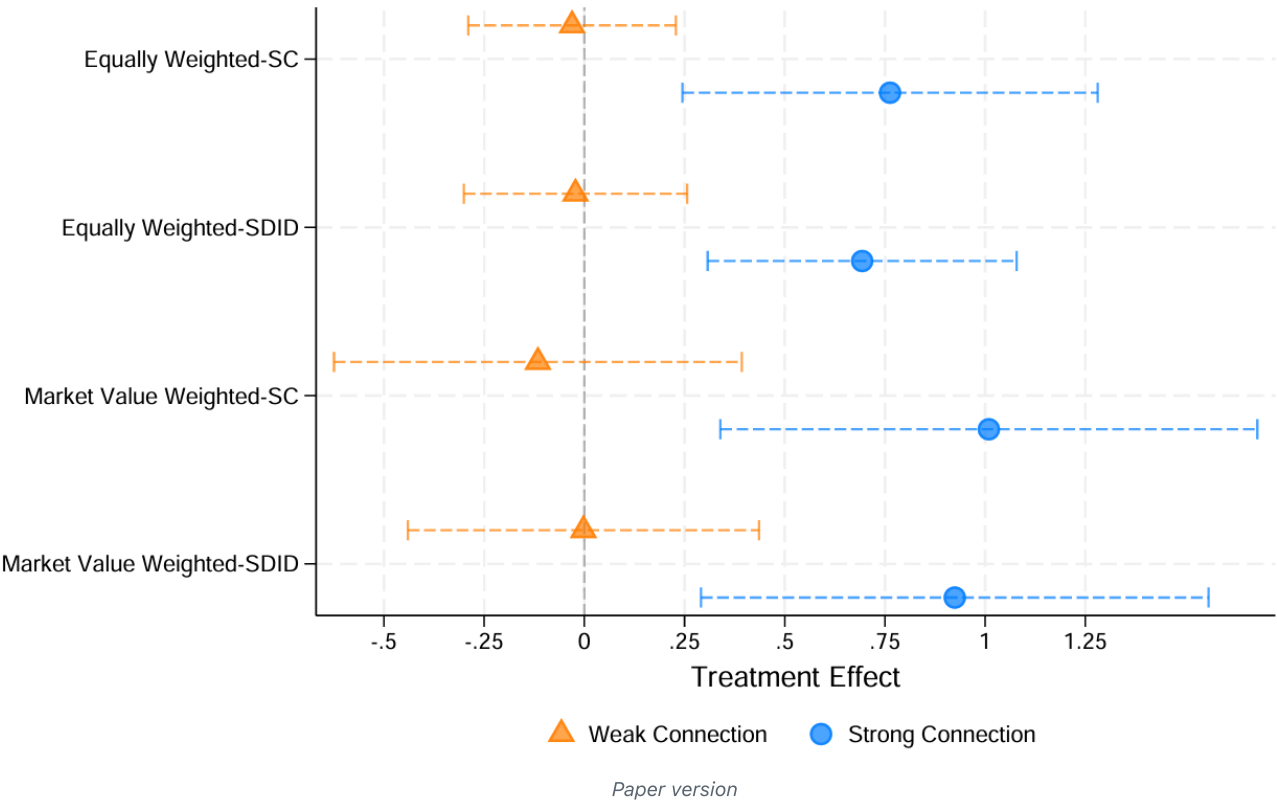


Figure 5: Figure A.13

- All other outputs are perfectly reproduced.

9.3 In-Text Numbers

- ☒ In-text numbers not verified.
- ☐ There are no in-text numbers, or all in-text numbers stem from tables and figures.

- ☐ There are in-text numbers, but they are not identified in the code

9.4 Other Issues

- Tables A.10 and A.11 are currently hard-coded, they should probably be produced by the code (see below):

Table A.10: Deadliest Mass Shootings in the US from 2010 to 2020

Location and Date	No. of Deaths
Aurora, CO, 7/12/2012	12
Newtown, CT, 12/14/2012	27
Washington, DC, 9/16/2013	12
San Bernardino, CA, 12/2/2015	14
Orlando, FL, 6/12/2016	49
Las Vegas, NV, 10/1/2017	58
Sutherland Springs, TX, 11/5/2017	25
Parkland, FL, 2/14/2018	17
Thousand Oaks, CA, 11/8/2018	12
Virginia Beach, VA, 5/31/2019	12
El Paso, TX, 8/2/2019	22

Table A.10

Table A.11: White Supremacist Incidents from the ADL

Location and Date	Attacker	No. of Deaths	Google News Articles
Overland Park, KS, 4/13/2014	Frazier Glenn Miller	3	443
Charleston, SC, 6/17/2015	Dylann Storm Roof	9	1090
New York, NY, 3/30/2017	James Harris Jackson	1	182
Charlottesville, VA, 8/12/2017	James Alex Fields	1	86,400
Lake Forest, CA, 1/2/2018	Samuel Woodward	1	145
Shawnee, KS, 7/2/2018	Ronald Lee Tidwell	1	4840
Pittsburgh, PA, 10/27/2018	Robert Gregory Bowers	11	372
Poway, CA, 4/27/2019	John T. Earnest	1	307

Table A.11

10 Classification

- ☐ full reproduction
- ☒ full reproduction with minor issues
- ☐ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☒ **Discrepancy in output** (either figures or numbers in tables or text differ)

- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements

10.2 Potential Personal Identifiable Information (PII)

△ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary:

- Data files with PII indicators: 131
- Variables flagged in data: 2713
- Code files with PII references: 98
- PII references in code: 923

10.2.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	18_APR_2022_PN_CatTermFreq.csv.gz	2	name, loc, location
Data	20_APR_2022_PN_CatTermFreq_Crime.csv.gz	6	name, loc, location, son, lon, sex
Data	21_SEP_2024_AW_CatTermFreq_v3.csv.gz	2	name, loc, location
Data	38651-0001-Data.csv.gz	14	name, city, zip, sex, child, lat, loc, house, community, second
Data	38651-0001-Data.dta	119	name, city, zip, son, lat, sex, address, child, community, school, loc, social, location, house, network, census, second
Data	ISS_table_ceo.csv.gz	2	son, country
Data	MPVDatasetDownload_feb2024.csv.gz	27	name, gender, url, address, street, city, zip, lat, son, census, house, lon, district
Data	MPVDatasetDownload_feb2024.xlsx	27	name, gender, url, address, street, city, zip, lat, son, census, house, lon, district

File Type	File	Variables/References	PII Categories
Data	MPV_GDELT_ORI_events_strict.csv.gz	18	country, lname, name, lat, lon, url
Data	PD_atlas_rosters.csv.gz	1	name
Data	PD_atlas_rosters.xlsx	1	name
Data	PD_atlas_rosters_Master_IACP.csv.gz	7	name, address, city, zip, phone, email
Data	black_mpv_xy_summer2020.csv.gz	98	lat, lon
Data	black_mpv_xy_summer2020.dta	98	lat, lon
Data	category_res_0.csv.gz	1	name
Data	category_res_0.dta	1	name
Data	category_res_1.csv.gz	1	name
Data	category_res_1.dta	1	name
Data	category_res_2.csv.gz	1	name
Data	category_res_2.dta	1	name

See [Appendix](#) for detailed listing of all flagged instances.